Sound and Waveforms

When a sound travels, the ________ ________ in the material it is travelling through. This means that they move forwards and backwards again and again.

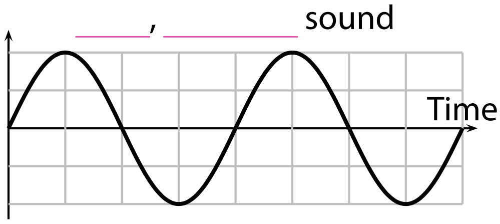
The time for each oscillation back and forth is called the __________.

When the sound reaches our ____ or a ___________, it makes a sensitive surface ________ too. This can be heard as sound.

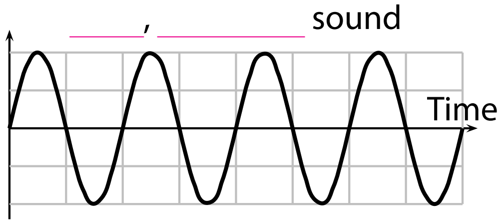
When a ___________ is connected to a computer, a ___________ or an ___________, the wave can be seen on a screen. The screen shows the ____________ changing as ____ passes.

The greatest displacement of a wave is called its _________.

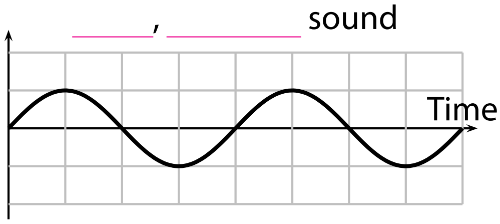
_____, ________ sound



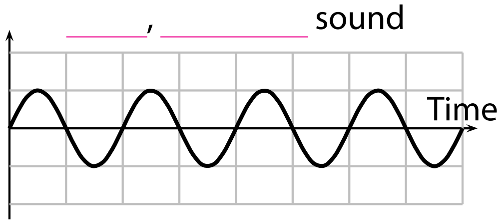
_____, _________ sound



_____, ________ sound



_____, _________ sound



1 Look at the waves shown above and answer the questions.

(a) Describe how the loud sounds' graphs differ from the quiet sounds' graphs.

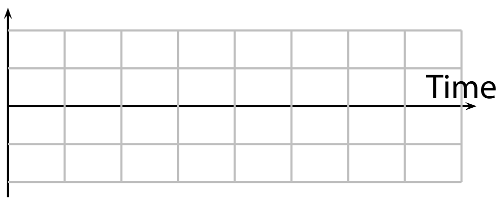
(b) Describe how the high sounds' graphs differ from the low sounds' graphs.

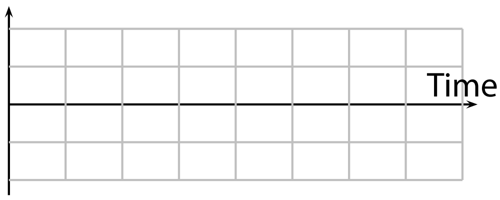
(c) Does a short time period mean that the sound is high pitched or low pitched?

(d) Does a large amplitude means that the sound is loud or quiet?

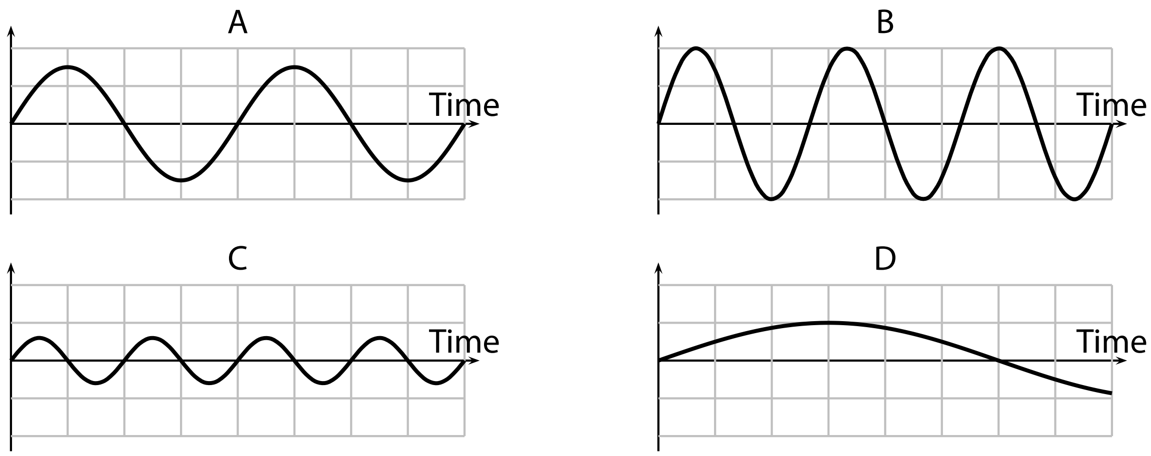
2 Draw waves on the graphs below to fit the descriptions.

(a) quieter and higher than the waves above (b) louder and lower than the waves above





3 Use these wave graphs to answer the questions below them.



(a) Put the letters ABCD in order to rank the waves from quietest to loudest.

(b) Put the letters ABCD in order to rank the waves from lowest to highest pitch.

(c) Which wave has the shortest time period?

The number of complete oscillations in one second is called the __________ of the wave.

Waves with a high frequency have a _____ time period and are ____________.

4 Which wave in question 3 has the lowest frequency?

5 Complete the sentences using **20 000 Hz**, **ultrasound**, **Hz**, **20 Hz**, **hertz**.

Frequencies are measured in __________ (symbol ________).

Our ears can hear sounds with frequencies between __________ and _____________.

Sounds above this range can not be heard and are called ____________.

Each oscillation takes a very short time so the times are measured in ____________ (___).

A millisecond is one ___________ of a second. There are _____ milliseconds in one second.

We write 1 s = _____ ms, or 1 ms = _____ s.

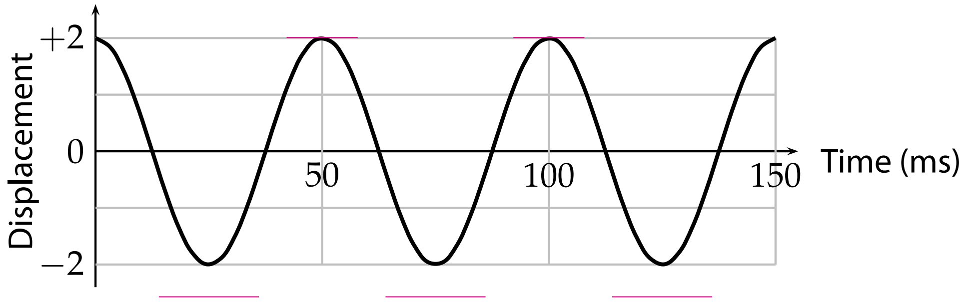
6 Write the times below in milliseconds.

(a) 0.002 s (b) 0.125 s (c) 0.02 s

7 Write the times below in seconds.

(a) 2000 ms (b) 3 ms (c) 40 ms

The **time period** of this wave is __ ms. This is the time from one ____ to the next ____.



8 What is the time period of this wave in seconds?

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9 In this question we look at the link between time period and frequency.

(a) A wave has a time period of 0.05 s. Work out how many times it oscillates each second using an equation.

1	$\div$	____	$=$	

(b) What is the frequency of a wave if its time period is 0.05 s?

(c) A wave has a time period of 0.002 s. Work out its frequency using an equation.

1	$\div$	time period (s)	$=$	frequency (Hz)
1	$\div$	____	$=$	

(d) Work out the frequency of a wave with a time period of 0.004 s using an equation.

1	$\div$	time period (s)	$=$	frequency (Hz)
1	$\div$	____	$=$	

10 A wave has a time period of 5 ms. Calculate its frequency.

11 What is the frequency of the wave in question 8?

A frequency of 4000 Hz is usually written ____. One kilohertz (1 kHz) is the same as ______.

12 Calculate the frequencies in kHz of waves with the following time periods.

(a) 10 ms

(b) 0.5 ms

(c) 40 ms

(d) 0.1 ms